



A robust approach to bioinformatics raw data processing

At Data Intuitive, we developed a **cutting-edge methodology and tool** for processing raw bioinformatics data.

Our mission is to streamline the development and maintenance of data workflows, minimising resource demands and maintenance burdens while prioritising quality, scalability, reproducibility, and effortless adaptability to evolving needs.

We have developed a state-of-the-art toolkit for developing workflows and our approach has been adopted by leaders in the industry.

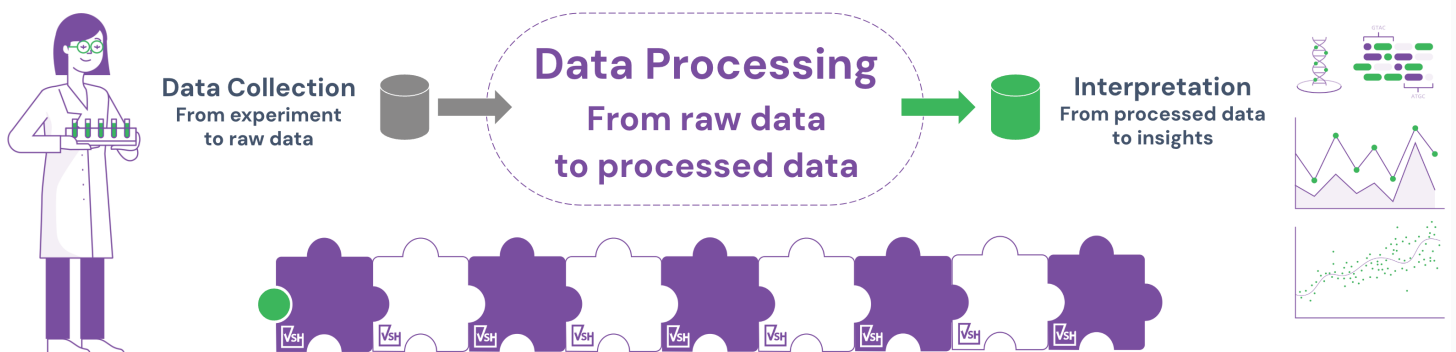
But first of all, why is there a need for such a toolkit? What are the common and current challenges within the industry? Let's walk through these together!

Common pitfalls and challenges with data workflows

In biotech, creating robust omics data workflows demands careful navigation of various challenges to ensure reliability and robustness. Below are ten common pitfalls that can undermine success:

- **Data quality:** Unreliable data invalidates results, fosters errors.
- **Reproducibility:** Irreproducible workflows erode credibility and compliance.
- **Scalability:** Poor scaling ability stalls projects under enterprise-level datasets.
- **Maintainability:** Unmaintainable badly designed workflows cause chaos.
- **Dependencies:** multiple dependencies can lead to unpredictable behavior.
- **Vendor lock-in:** Vendor reliance limits flexibility and adaptability shifting platforms..
- **Efficiency & collaboration:** Poor efficiency wastes research time and IT expert time.
- **Multi-platform processing:** Platform inconsistency creates operational inefficiencies.
- **Security & compliance:** Weak security jeopardises compliance and confidence.
- **Version control:** Lacking version control fosters mismatched code, confusion.

For more information on these pitfalls, click here to receive our article
"10 Common pitfalls in omics data processing—and how to avoid them"





Key advantages of adopting our approach

Reduce costs and simplify operations

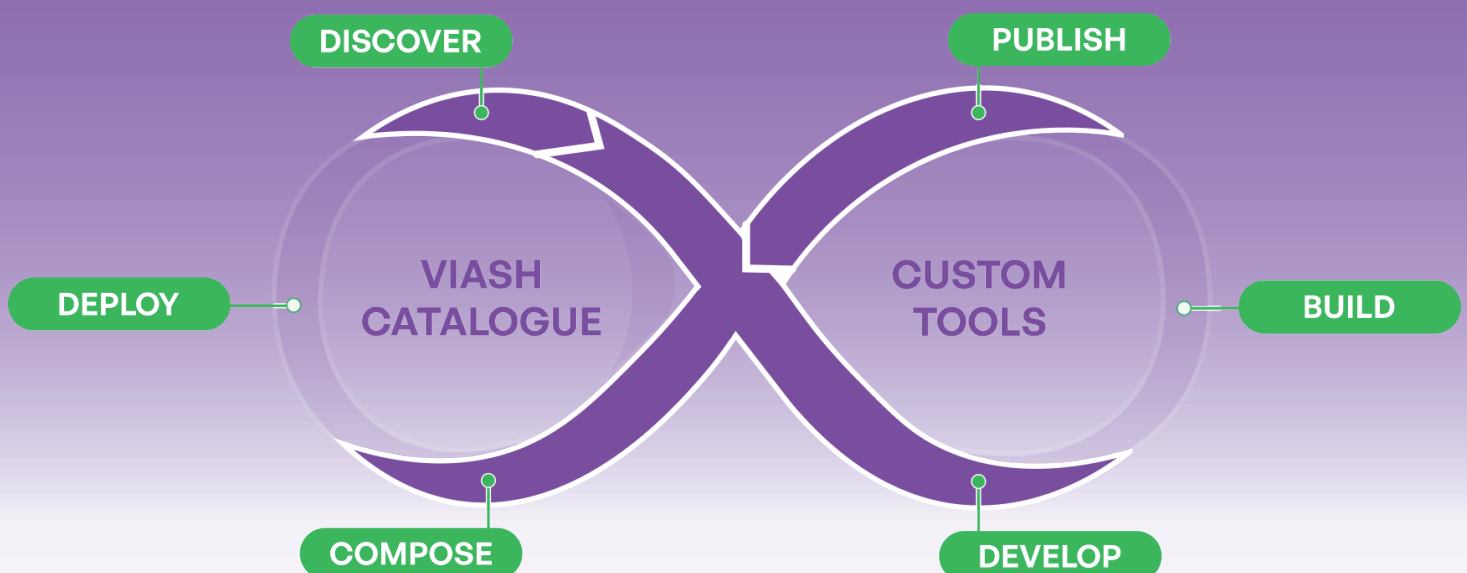
- **Reduced Maintenance Overhead:**
Our clients consistently report long-term cost savings and reduced dependency on niche expertise.
- **Streamlined Operations:**
By separating scientific processes from IT operations, we optimise resource allocation and eliminate unnecessary bottlenecks.
- **Extensive Catalogue of Building Blocks:**
Leverage our large library of pre-built modular building blocks featuring the most common bioinformatics tools, ready to integrate seamlessly into your workflows and cutting workflow development time by 50% or more.

Rapid adaptation to change

- **Effortless Tool Updates:**
Our modular design enables precise updates to individual components, avoiding the need for complete workflow overhauls when integrating new tools or technologies.
- **Future-Proof Architecture:**
Maintain flexibility and adaptability with a design that avoids vendor lock-in and proprietary platform constraints, ensuring long-term resilience in a rapidly evolving industry.

Scalable, platform-independent workflows

- **Reliable, Reproducible Results:**
Automatically track and manage tool and workflow versions over time, ensuring updates are monitored and consistency is maintained across projects.
- **Multi-platform flexibility:**
Each modular component is self-contained with all necessary dependencies, enabling seamless execution on local servers, HPC clusters, or cloud environments—no specialised setup required.
- **Ready-to-Use Solutions:**
Built for immediate scalability, our workflows handle high-volume or complex data projects effortlessly, thanks to its "batteries-included" approach.

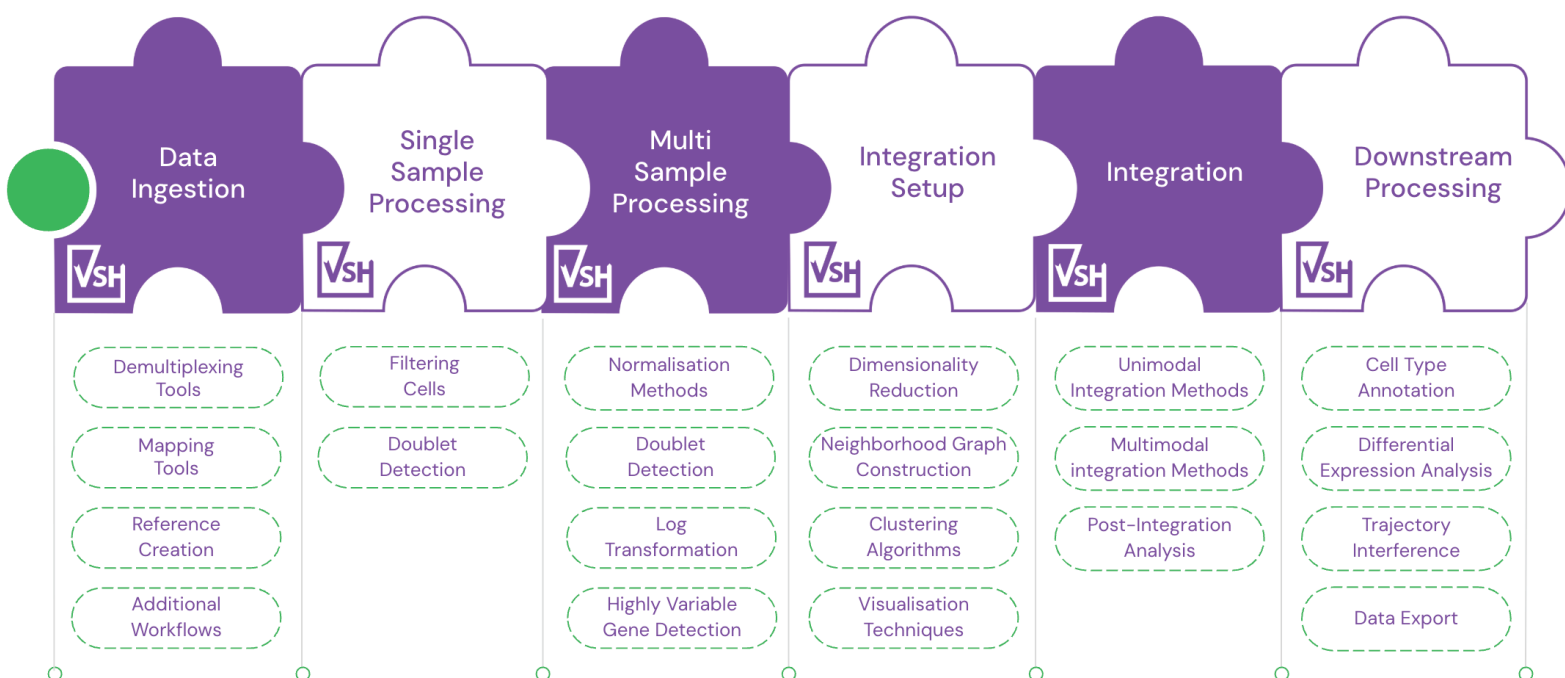




Introducing Viash & Viash Catalogue: our core tooling ecosystem

- At the heart of our workflow development is Viash, a **next-generation solution designed to simplify and streamline the creation of data processing pipelines**. Viash enables organisations to build workflows quickly and reliably within a unified framework. Instead of manually writing and assembling complex scripts, Viash automatically generates self-contained building blocks that include all the requirements a computational biotech tool needs—such as inputs, outputs, and software dependencies.
- These **ready-to-use building blocks** integrate seamlessly into pipelines that are efficient, scalable, and easy to maintain. Viash supports multiple programming languages and works effortlessly with your existing environments, ensuring adaptability while allowing you to continue using your preferred tools in an organised and scalable manner.
- Building on Viash, we have developed an **extensive, open-source catalogue of modular workflows**. This catalogue allows us to create custom solutions by complementing existing proven building blocks with additional modules tailored to unique project requirements—without the high costs typically associated with development and maintenance. This modular approach enables us to deliver tailored, efficient solutions faster, empowering clients to focus on their core objectives while we handle the complexities of workflow development.

Example workflow Viash Catalogue: single cell multi omics processing





Data Intuitive

Founded in 2011, Data Intuitive is a Belgian-based bioinformatics company specialising in data workflow development. Since transitioning to startup mode in 2021, we have leveraged a decade of expertise to create innovative solutions that simplify and optimise data processing workflows.

We believe in clear, transparent communication, enabling us to build trust with our clients while delivering long-term, quality-driven solutions that address current challenges and anticipate future developments in biotech data processing.

Our vision is to lead the biotech sector in redefining data workflow approaches, with an emphasis on sustainability, efficiency, and long-term optimisation.

What Sets Us Apart

- **Accelerated Discovery:**
We streamline data workflows, enabling faster, cost-effective experimentation and analysis to help clients shorten the timeline from experiment to insight.
- **Open-Source Advocacy:**
Our commitment to open-source tools and resources reflects our belief in the power of collaborative, forward-moving science.
- **Innovation-Driven Culture:**
We foster a dynamic environment where IT experts push technical boundaries, driving growth through cutting-edge technology rather than temporary solutions.
- **Robust Foundations:**
By building on industry-standard computer science best practices, we deliver stable, maintainability and future proof systems.

Beyond Traditional Consultancy

- At Data Intuitive, we go beyond the typical consultancy model by focusing on crafting solutions that endure. Our multidisciplinary team of experienced IT professionals, data scientists, and computational biologists, brings deep expertise in biotech data workflows.
- Unlike traditional consulting firms, we've developed our own tooling ecosystem, including Viash and the Viash Catalogue, which empowers clients with robust, adaptable solutions that last, rather than creating dependency on external support.
- We work closely with clients to provide flexible, solution-focused strategies tailored to their needs. By deploying the latest technologies and ensuring consistent support, we deliver outcomes that transcend traditional consultancy services.